

Math 1314 Lesson 23

1) Find the zeros of the polynomial function $f(x) = x^2 - 7x + 6$

a) Graph $f(x) = x^2 - 7x + 6$. x -intercepts = ? and ?

b) Factor $x^2 - 7x + 6 = (\text{ ? })(\text{ ? })$

c) The zeros of the function $f(x) = x^2 - 7x + 6$ are ? and ?

2) Find the zeros of the polynomial function $f(x) = 4x^3 - 76x + 120$.

a) Graph $f(x) = 4x^3 - 76x + 120$. x -intercepts = ?, ? and ?

b) Factor $4x^3 - 76x + 120 = (\text{ } ? \text{ })(\text{ } ? \text{ })(\text{ } ? \text{ })$

c) The zeros of the function $f(x) = 4x^3 - 76x + 120$ are ?, ? and ?

3) Find the zeros of the polynomial function $f(x) = 2x^3 - x^2 - 27x + 36$.

a) Graph $f(x) = 2x^3 - x^2 - 27x + 36$. x -intercepts = ?, ? and ?

b) Factor $2x^3 - x^2 - 27x + 36 = (\text{?})(\text{?})(\text{?})$

c) The zeros of the function $f(x) = 2x^3 - x^2 - 27x + 36$ are ?, ? and ?

4) Find the zeros of the polynomial function $f(x) = 4x^3 - 8x^2 - 17x + 30$.

a) Graph $f(x) = 4x^3 - 8x^2 - 17x + 30$. x -intercepts = ?, ? and ?

b) Factor $4x^3 - 8x^2 - 17x + 30 = (\text{ } ? \text{ })(\text{ } ? \text{ })(\text{ } ? \text{ })$

c) The zeros of the function $f(x) = 4x^3 - 8x^2 - 17x + 30$ are ?, ? and ?

5) Find the zeros of the polynomial function $f(x) = x^4 - 2x^3 - 5x^2 - 2x + 24$

a) Graph $f(x) = x^4 - 2x^3 - 5x^2 - 2x + 24$. x -intercepts = ? and ?

b) From part (a), two of the factors of $4x^3 - 8x^2 - 17x + 30$ are (factor 1 ?)(factor 2 ?)

c) Use synthetic division to divide $4x^3 - 8x^2 - 17x + 30$ by factor 1 and factor 2.

d) Set the result in part (c) equal to 0 and solve.

Hint: Use the Quadratic Formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

e) The zeros of the function $f(x) = x^4 - 2x^3 - 5x^2 - 2x + 24$ are ?, ?, ? and ?

6) Find the zeros of the polynomial function $f(x) = 2x^4 - 5x^3 - 13x^2 + 16$

a) Graph $f(x) = 2x^4 - 5x^3 - 13x^2 + 16$. x -intercepts = ? and ?

b) From part (a), two of the factors of $2x^4 - 5x^3 - 13x^2 + 16$ are (factor 1 ?)(factor 2 ?)

c) Use synthetic division to divide $2x^4 - 5x^3 - 13x^2 + 16$ by factor 1 and factor 2.

d) Set the result in part (c) equal to 0 and solve.

Hint: Use the Quadratic Formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

e) The zeros of the function $f(x) = 2x^4 - 5x^3 - 13x^2 + 16$ are ?, ?, ? and ?

